

Contribution of Each Member

1. V. RUTHVIKA (2500090173)

Focus:

System design, architecture, and core AQMAS functionality

Work:

- Designed the overall structure and workflow of the Air Quality Monitoring and Analysis System
- Developed the main modules for collecting and processing air quality data
- Implemented core logic for calculating AQI (Air Quality Index)
- Ensured proper validation of input data from sensors/datasets
- Coordinated with team members for integration of all components

Contribution:

- Built a strong foundation for the AQMAS project
 - Ensured smooth functioning of the system
 - Improved accuracy in AQI calculations
 - Played a key role in integrating all modules
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2. P. SAI TEJA (2500090281)

Focus:

Data handling, storage, and processing

Work:

- Implemented data storage using CSV files and Panda's library
- Developed functions to store and retrieve air quality data
- Cleaned and reprocessed environmental datasets
- Handled missing and incorrect values in data
- Ensured consistency and proper formatting of records

Contribution:

- Enabled efficient data storage and retrieval
- Improved data reliability and accuracy
- Reduced errors in data processing
- Supported smooth data flow across modules

3. CH. SIVANI (2500090250)

Focus:

Data analysis and AQI computation

Work:

- Implemented AQI calculation using standard formulas
- Performed statistical analysis using NumPy
- Calculated average, maximum, and minimum pollutant levels
- Identified patterns and trends in air quality data
- Assisted in validating analytical outputs

Contribution:

- Added analytical capabilities to the system
 - Helped in understanding pollution trends
 - Improved decision-making using data insights
 - Enhanced overall accuracy of analysis
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4. Y. LOCHANA (2500090305)**Focus:**

Visualization, interface, and reporting

Work:

- Created graphs and charts using Matplotlib (line charts, bar graphs)
- Designed user-friendly interface for displaying AQI results
- Generated visual reports for air quality analysis
- Improved readability and presentation of outputs
- Assisted in preparing final project documentation

Contribution:

- Made the system interactive and easy to understand
- Simplified data interpretation using visualizations
- Enhanced presentation quality of the project
- Improved overall user experience